

## Evidence sheet

0.9329

CV AUC

0.84

THRESHOLD

0.921

PRECISION @ 20%  
PREVALENCE

0.641

EXPECTED RECALL

None

ROWS DELETED

## 1 What we cleaned and why (top 5)

1. Numbers stored as text: 9900 decimal commas, 1389 thousands separators, 3256 missing tokens (?, --, unknown); 4 column names and a 2-line banner.
2. Scale slips: 610 cells  $\times 100$  and 613  $\times 1000$  (factor chosen by the sister columns); 409 float-junk  $1e-11$  values set to 0.
3. 40 negative areas: `abs()` still disagreed with the radius, so they were rebuilt, not sign-flipped.
4. 2601 missing radius/perimeter/area cells rebuilt from geometry ( $P=6.479R$ ,  $A=3.080R^2$ , learned on train). Missing size cells went from 2654 to 99; no rows dropped.
5. Labels in 67 raw spellings mapped to M/B. 4328 IDs normalised (ID-, commas, .0, tabs), which is what reveals the duplicates. 9 ages and 79 dates set missing.

## 2 Problems that were not obvious (between columns)

- **Perimeter and area swapped** in 158 rows ( $P > A$  is impossible): swapped back. **Radius in cm** in 1378 cells ( $P/R = 65$ , not 6.5):  $\times 10$ . Rows with  $P/R$  outside [5, 8]: 10.4%  $\rightarrow$  0.00%.
- **346 patients entered twice, 50 with opposite diagnoses.** Merged; 48 labels repaired by the follow-up code (97.7% agreement on uncontested rows), the rest soft-labelled.
- **Training labels are noisy:** 0.044 / 0.057 of the most confident 20% carry the opposite label. **Site C** reads larger even in benign cases (radius  $\times 1.073$ ).

## 3 Columns we dropped, and why

- **biopsy\_followup\_code = target leakage:** ONC-REF is 96.1% malignant and ROUTINE 2.1%; on its own it gives AUC 0.971. It is written after the diagnosis, so it does not exist at prediction time.
- notes (malignant rate 34%-43% in every template), sample\_ref (random lab string), IDs and dates: no signal. center: a calibration risk.

## 4 How we validated

- StratifiedGroupKFold, 5 folds  $\times$  3 repeats, grouped by the *normalised* patient\_id after merging duplicates. The code asserts that no patient lands in both train and validation. Repair parameters are learned on train only.
- Ensemble of logistic regression and gradient boosting: CV AUC  $0.9329 \pm 0.0004$ , scored against noisy labels. Repairs raise the linear model from AUC 0.9038 to 0.9229 and recall@P90 from 0.5473 to 0.7613.

## 5 Our threshold, and why

- **$t = 0.84$ .** It is the lowest  $t$  where precision stays  $\geq 0.92$  after removing label noise *and* halving prevalence to 20% (the brief warns that test is drawn differently), and where precision on the labels as given is  $\geq 0.90$  (0.933; bootstrap 5th percentile 0.924).
- Expected precision 0.921 under that stress, recall 0.641. The default  $t = 0.5$  would give only 0.812.
- Cost: dropping  $t$  by 0.10 costs 0.14 false alarms per extra cancer at the training mix (0.34 at 20% prevalence) and breaks the stressed floor (0.887).

## 6 What would break this model

- Younger women ( $<40$ : CV AUC 0.888 vs 0.9329 overall), a new site or scanner with its own calibration, or an unseen unit convention.
- A population with far fewer than 20% cancers (screening): precision would drop below 0.90, so the threshold must be re-set.
- **Never** a stand-alone diagnosis and never a reason to skip biopsy. It is a triage aid on top of a pathologist.